Benjamin Chatman

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EDUCATION

Michigan State University, East Lansing, MI May 2028

- *Bachelor of Science in* **Computer Science**
- *Bachelor of Arts in* **Mathematics**
- GPA: 3.15/4.00

WORK EXPERIENCE

North Peak Leads, *Software Engineer Intern,* Grand Rapids, MI July 2026 – Present

- Built a website to intake new clients and gather information from them to give them a quote and to allow for follow up information.
- Built a small functional database from scratch using SQL to keep track of client names, appointment times, and other crucial information.

PROJECTS & LEADERSHIP EXPERIENCE

Chess GUI and Engine | C++

- A chess GUI capable of handling all chess moves in the standard chess rule book.
- Utilizes alpha-beta pruning and quiescence search methods to search for the best moves under a given time constraint.
- Competes at an elo rating of 1700 on a 3 second move timer.

Planar | C++

- A multi-user music app allowing songs to be listened to together across devices.
- Delay between devices is below 15ms, about 98% of the time.
- Written using Youtube's API, allowing any song on Youtube to be played in app.

3D Renderer | C++

- A raycast engine akin to older video games such as DOOM and Wolfenstein 3D.
- Uses a 2D array as a map to calculate distance between the camera and the objects to render them onto the screen of the user.
- Optimized and concise code capable of running 2000+ frames per second on low-end hardware.

FRC Team 3357, *Lead Programmer*, Grand Rapids, MI May 2022 – August 2024

- Led a top 100 team in the country to win the world championship multiple times.
- High-level object-oriented C++ programming which allowed the robot to function autonomously and manually.
- Teach new students the C++ programming language and our meticulous methodology of working through new problems every year, to ensure the success of themselves and the team.
- Experience with Git to keep code organized and sectioned so multiple students can work simultaneously.

TECHNICAL SKILLS

- *Computer Languages:* Python, C++, Java, C, SQL
- *Tools:* Git, Microsoft Office, wxWidgets, Visual Paradigm
- *Skills:* Unix/Linux Operating Systems, Codebase Management, UML Design